BIG DATA BY AMRITA KESHARI

Unit 3

Introduction to Hive

Hive is a Data Warehousing tool that sits on top of Hadoop.Hive is used to process structured data in Hadoop. The three main tasks performed by Apache Hive are:
1. Summarization
2. Querying
3. Analysis
Facebook initially created Hive component to manage their ever-growing volumes of log data. Later Apache software foundation developed it as open-source and it came to be known as Apache Hive. Hive makes use of the following:
1. HDFS for Storage.
2. MapReduce for execution.
3. Stores metadata/schemas in an RDBMS.
Hive provides HQL (Hive Query Language) or HiveQL which is similar to SQL. Hive compiles SQL queries into MapReduce jobs and then runs the job in the Hadoop Cluster. It is designed to support OLAP (Online Analytical Processing). Hive provides extensive data type functions and formats for data summarization and analysis.

Hive Features

1. It is similar to SQL.
2. HQL is easy to code.
3. Hive supports rich data types such as structs, lists and maps.
4. Hive supports SQL filters, group-by and order-by clauses.
5. Custom Types, Custom Functions can be defined.

Hive Data Units

1. Databases: The namespace for tables.
2. Tables: Set of records that have similar schema.
3. Partitions: Logical separations of data based on classification of given information as per specific at tributes. Once hive has partitioned the data based on a specified key, it starts to assemble the records into specific folders as and when the records are inserted.
4. Buckets (or Clusters): Similar to partitions but uses hash function to segregate data and determines the cluster or bucket into which the record should be placed.

HIVE ARCHITECTURE

Hive Architecture is depicted in Figure 9.7. The various parts are as follows:
1. Hive Command-Line Interface (Hive CLI): The most commonly used interface to interact with Hive.
2. Hive Web Interface: It is a simple Graphic User Interface to interact with Hive and to execute query.
3. Hive Server: This is an optional server. This can be used to submit Hive Jobs from a remote client.
4. JDBC/ODBC: Jobs can be submitted froma JDBC Client. One can write a Java code to connect to Hive and submit jobs on it.

5. Driver: Hive queries are sent to the driver for compilation, optimization and execution.
6. Metastore: Hive table definitions and mappings to the data are stored in a Metastore. A Metastore consists of the following:
• Metastore service: Offers interface to the Hive.
• Database: Stores data definitions, mappings to the data and others.

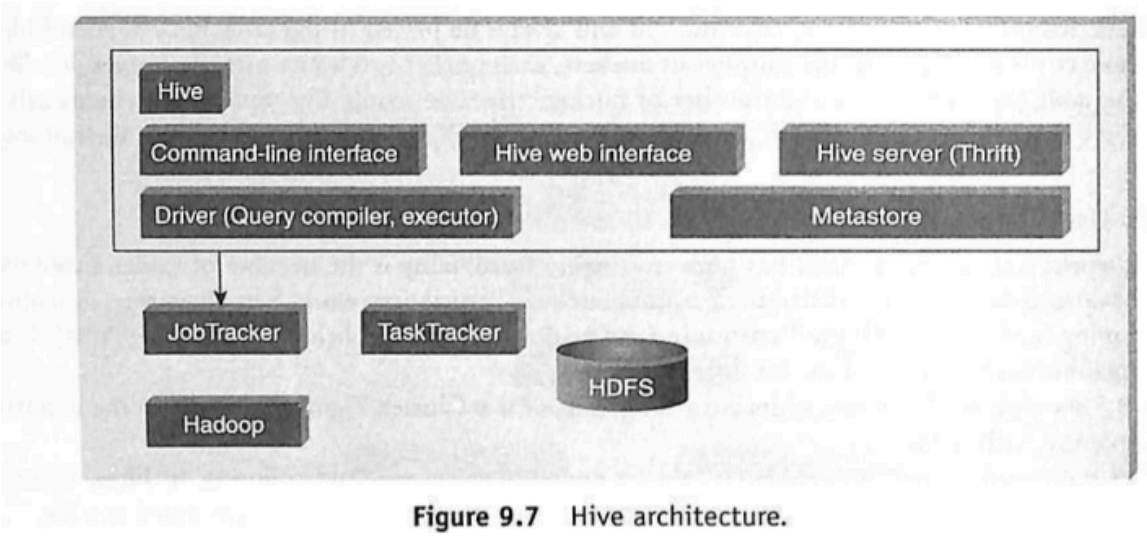


Figure 9.7 Hive architecture.

HIVE DATA TYPES

Primitive Data Types

Numeric Data Type	
TINYINT	1-byte signed integer
SMALLINT	2-byte signed integer
INT	4-byte signed integer
BIGINT	8-byte signed integer
FLOAT	4-byte single-precision floating-point
DOUBLE	8-byte double-precision floating-point number
String Types	
STRING	
VARCHAR	Only available starting with Hive 0.12.0
CHAR	Only available starting with Hive 0.13.0
Strings can be expressed in either single quotes (') or double quotes (")	
Miscellaneous Types	
BOOLEAN	
BINARY	Only available starting with Hive

9.3.2 Collection Data Types

Collection Data Types	
STRUCT	Similar to 'C' struct. Fields are accessed using dot notation. E.g.: struct('John', 'Doe')
MAP	A collection of key–value pairs. Fields are accessed using [] notation. E.g.: map('first', 'John', 'last', 'Doe')
ARRAY	Ordered sequence of same types. Fields are accessed using array index. E.g.: array('John', 'Doe')

HIVE QUERY LANGUAGE (HOL)

Hive query language provides basic SQL like operations. Here are few of the tasks which HQL can do easily.

1. Create and manage tables and partitions.
2. Support various Relational, Arithmetic, and Logical Operators.
3. Evaluate functions.
4. Download the contents of a table to a local directory or result of queries to HDFS directory.

1. DDL (Data Definition Language) Statements

These statements are used to build and modify the tables and other objects in the database.
The DDL commands are as follows:

1. Create/Drop/Alter Database
2. Create/Drop/Truncate Table
3. Alter Table/Partition/Column
4. Create/Drop/Alter View
5. Create/Drop/Alter Index
6. Show 7. Describe

2. DML (Data Manipulation Language) Statements

These statements are used to retrieve, store, modify, delete, and update data in database.
The DML commands are as follows:

1. Loading files into table.
2. Inserting data into Hive Tables from queries.
3. Delete and update.

WHAT IS PIG?

Apache Pig is a platform for data analysis. It is an alternative to MapReduce Programming. Pig was developed as a research project at Yahoo. Key Features of Pig

1. It provides an engine for executing data flows (how your data should flow). Pig processes data in parallel on the Hadoop cluster.
2. It provides a language called "Pig Latin" to express data flows.
3. Pig Latin contains operators for many of the traditional data operations such as join, filter, sort, etc.
4. It allows users to develop their own functions (User Defined Functions) for reading, processing, and writing data.

THE ANATOMY OF PIG

The main components of Pig are as follows:

1. Data flow language (Pig Latin).
2. Interactive shell where you can type Pig Latin statements (Grunt).
3. Pig interpreter and execution engine.

Refer Figure 10.1.

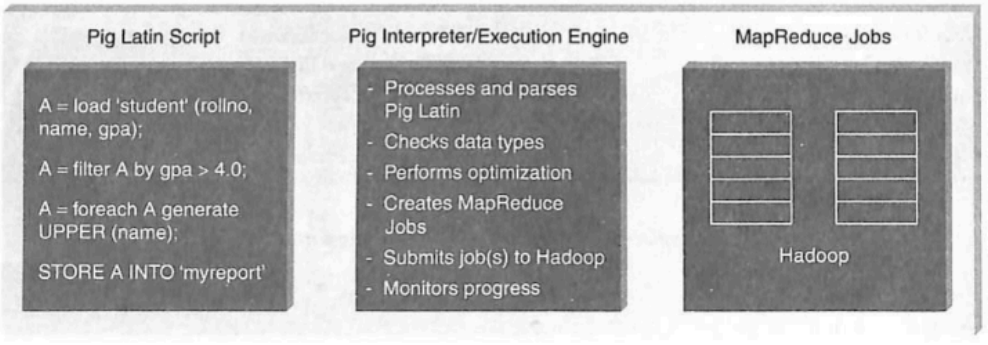


Figure 10.1 The anatomy of Pig.

PIG ON HADOOP

Pig runs on Hadoop. Pig uses both Hadoop Distributed File System and MapReduce Programming. By default, Pig reads input files from HDFS. Pig stores the intermediate data (data produced by MapReduce jobs) and the output in HDFS. However, Pig can also read input from and place output to other sources. Pig supports the following:

1. HDFS commands.
2. UNIX shell commands.
3. Relational operators.
4. Positional parameters.
5. Common mathematical functions.
6. Custom functions.
7. Complex data structures

PIG PHILOSOPHY

Figure 10.2 describes the Pig philosophy.

1. Pigs Eat Anything: Pig can process different kinds of data such as structured and unstructured data.
2. Pigs Live Anywhere: Pig not only processes files in HDFS, it also processes files in other sources such as files in the local file system.
3. Pigs are Domestic Animals: Pig allows you to develop user-defined functions and the same can be included in the script for complex operations.
4. Pigs Fly: Pig processes data quickly.

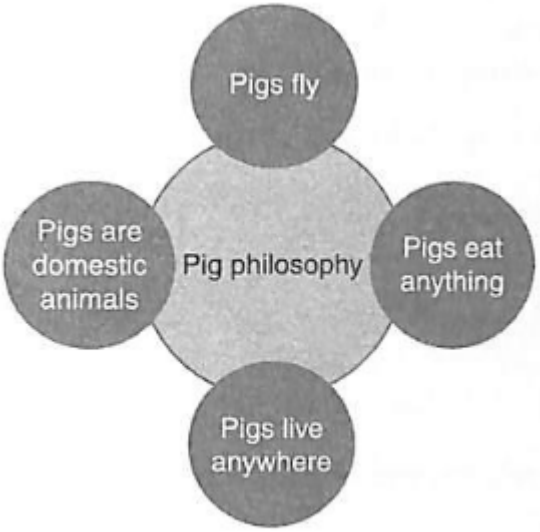


Figure 10.2 Pig philosophy.

USE CASE FOR PIG: ETL PROCESSING

Pig is widely used for "ETL" (Extract, Transform, and Load). Pig can extract data from different sources such as ERP, Accounting, Flat Files, etc. Pig then makes use of various operators to perform transformation on the data and subsequently loads it into the data warehouse. Refer Figure 10.3.

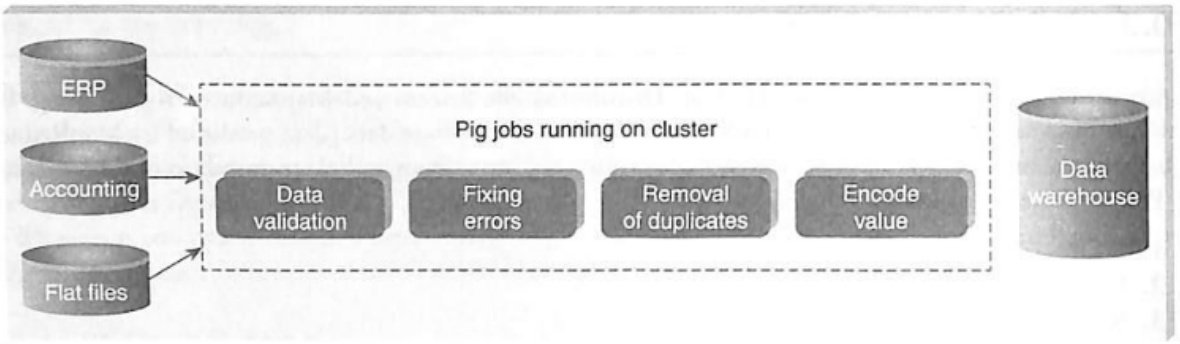


Figure 10.3 Pig: ETL Processing.

DATA TYPES IN PIG

Simple and complex

Table 10.3 Simple data types supported in Pig

Name	Description
Int	Whole numbers
Long	Large whole numbers
Float	Decimals
Double	Very precise decimals
Chararray	Text strings
Bytearray	Raw bytes
Datetime	Datetime
Boolean	true or false

Table 10.4 Complex data types in Pig

Name	Description
Tuple	An ordered set of fields. **Example: (2,3)**
Bag	A collection of tuples. **Example: {(2,3),(7,5)}**
map	key, value pair **(open # Apache)**

RUNNING PIG

You can run Pig in two ways:

1. Interactive Mode.

You can run Pig in interactive mode by invoking grunt shell. Type pig to get grunt shell,Once you get the grunt prompt, you can type the Pig Latin statement

2. Batch Mode.
You need to create "Pig Script" to run pig in batch mode. Write Pig Latin statements in a file and save it with .pig extension.